## P1. Assignment 1.1 Linux installation and Administration

Linux Installation and Administrative Tasks

Objective

This assignment is intended to familiarize students with Linux administration, covering security aspects of installation and performing several administration tasks in a Linux Virtual Machine. You may need to install different tools like asciinema, curl, nano, etc. Please make sure to do so, and keep track of the activities performed in your assignment report.

Deliverable for this assignment will consist of two parts: you will record an Asciinema .cast file, and also prepare a report with screenshots and explanation of the process you have followed.

Assignment Phases

The assignment is divided into two phases, which are required to be completed in order. Phase 1 is intended for the installation of a Linux virtual machine, followed by Phase 2 where students will perform several administrative tasks.

**Phase 1: Linux installation**

LinuxLinks to an external site. is available in many different flavours, often referred to as distributionsLinks to an external site., which may have varied choices when it comes to various user-space software and kernel configurationLinks to an external site. parameters. These distributions comes with various patches which can be leveraged by their users to enable features that may not be supported upstreamLinks to an external site..

While there are myriad of Linux flavours available, for the purposes required by this course, we'd recommend some popular and well-supported distros that you might encounter in the enterprise environment.

| Distro Name | Long Term Support (LTS) version | Associated Company | Package Manager | Default MAC Implementation |
|---|---|---|---|---|
| Ubuntu Server (24.04)Links to an external site. | Ubuntu Server (24.04)Links to an external site. | CanonicalLinks to an external site. | AptLinks to an external site. | AppArmorLinks to an external site. |

As you can see, we recommend a server version, that ships WITHOUT a Graphical User Interface (GUI). This is because we want you to be familiar with the command line interface, which will always be available, and not to mention, this version comes with a comparatively lightweight deployment and lesser overhead. If you prefer to install a Desktop version, that is fine, but keep in mind that it will take double the space than the server version.

Make sure you have at least 10GB hard disk space and 4 GB RAM on your system. This is the bare minimum, and having more RAM and disk space will be allow a smoother experience.

**Task 1 - Install VirtualBox**

Oracle VM VirtualBoxLinks to an external site. is a free and open-source hosted hypervisorLinks to an external site. for x86 computers and is under development by Oracle Corporation. We will be using it to run multiple virtual machinesLinks to an external site. that will be used throughout this course. Download and install VirtualBox for your host OS using this linkLinks to an external site..

For MacOS users, UTMLinks to an external site. or VMWARELinks to an external site. or ParallelsLinks to an external site. can be used to run ubuntu OS. UTM is an Apple-only offering and leverages the Apple native GUI. For this Lab, We will be using UTM as it offers better performance. Download and install UTM for your host OS using this linkLinks to an external site..

**Task 2 - Install Ubuntu**

Download Ubuntu Server (24.04)Links to an external site.. If you have a Apple M1, M2 processor, make sure you choose the correct image file according to your system's architectureLinks to an external site. (likely to be AMD64 / x86_64).

For MacOS users, Ubuntu iso file can be downloaded using this link.Links to an external site. Installation guide for UTM is available in this link.Links to an external site.

Once the ISO image file has been downloaded, use the following steps to install Ubuntu:

- Open VirtualBox and click on **New** to Create a new Virtual Machine.
- **Name and OS:** Enter a name, and set Type -> **Linux** and Version -> **Ubuntu 32/64** depending on your system's architecture.
- **Memory Size:** Set the RAM to 2GB.
- **Hard Disk:** Select **Create a new hard disk now**.
- **Hard disk file type:** Choose **VDI** (which is Oracle's own format).
- **Storage on physical hard disk:** Dynamic Size.
- **File location and size:** Enter an appropriate name and choose at least **20 GB** for the hard disk size.

This creates a new Virtual Machine. Remember right now we just have a virtualized hard disk without any OS. The next step is to install Ubuntu. Following are the steps to install Ubuntu Minimal:

- Open the new VM's settings by click **Settings** from top-right menu bar and navigate to **Storage**.
- Under **Storage Devices** you will see **Controller: IDE** (port used for CD/DVD). Click on **Empty** under **Controller: IDE** and click on **Disk** shown on right side.
- From the popup menu, choose **Choose Virtual Optical Disk File** and select the downloaded ISO file 'mini.iso'.
- Click **OK** and start the VM by click **Start** from top-right menu bar.

Once the VM is booted use the following steps to install. Please look into them carefully and check with the instructor or the TA if you have questions, some of these topics may appear in our class quizzes (i.e.: password selection, disk encryption, security updates, etc.)

- Enter **Install**.
- Choose **language**, **country** and **keyboard** settings.
- Enter a **hostname** of your choice (after enter your hostname, press 'Tab' key from your keyboard to direct to button) and choose an ubuntu archive mirror that is closest to your country. This will reduce the latency in downloading system updates.
- Leave the proxy information blank as we are not configuring any proxies right now.
- Create default user account by entering your name and an appropriate username.
- Make sure you create a strong passwordLinks to an external site., as it is important to protect your account from brute-force attacks. If you want to see your password, use 'Tab' key to point 'Show Password in Clear' and press 'Space' key to check the box.
- Choose the correct time zone.
- **Partition disks:** It is a good practice (and more secure) to enable full disk encryption (FDE)Links to an external site..
  Make sure you use the FDE option to **use entire disk and set up encrypted LVM**.

**Note:** If you lose your disk encryption passphrase, the VM will become **inaccessible** and its data **will not** be recoverable. This encryption passphrase should be stored securely (in your password manager).

- Choose the disk to partition, which is the virtual hard disk in this case.
- Click Yes to write changes to the disk and configure LVM.
- Choose a strong passphrase to be used for Full Disk Encryption (FDE).
- Choose maximum disk space as shown below: -
- **PAM Configuration:** Select "install security updates automatically".
- **Software Selection:** Make sure you select OpenSSH at the software selection screen by pressing spacebar key as it will be required to SSH into the machine. This step can also be performed once the OS is booted.
- **Install Grub:** Choose to install GRUB bootloader to the master boot record as ubuntu is the only OS present on the system (VM).
- **System Clock:** Check your OS's default system clock is set to UTC and select yes/no accordingly.
- To boot into the OS, first you need to remove the disk (.iso) that was mounted to the VM. Power off the VM and go to VM's settings and remove the storage media. Once the media has been removed, then start the VM again to log into the minimal Ubuntu installation.

Deliverables for Phase 1:

**Screenshot 1.1** showing the prompt for the Full Disk Encryption password on first boot., **Screenshot 1.2** showing the booted VM after a successful login with the user account created during the installation.

**Phase 2: Perform Administration Tasks**

This phase is intended for you to complete several tasks in relation with the security of the Linux system. Keep in mind that to complete the lab you need to record your operations with the asciinema CLI recording. You also need to keep track of what you do in your assignment report, with the screenshots requested. It is mandatory to properly tag each screenshot in your report to identify which task it is related to.

**Task 1 - Install and start asciinema**

Type carefully the following command and substitute your **Your NU Name** with the user of your northeastern email account. For example if your email is smith.m@norhteastern.edu, your name will be **smith.m**. Store the cast file into your linux VM home directory

- 
    - 
    - $ asciinema rec /home/user/YOUR_NU_NAME.cast
    - ~ Asciicast recording started.
    - ~ Hit Ctrl-D or type "exit" to finish.
    - $

**Task 2- Create Users**

Create a two new users named your_nu_name_admin1 and your_nu_name_admin2 with password neulinux (for both). Also create a user named your_nu_name_regular with password newlinux.

**NOTE: Please ensure to substitute "your_nu_name" with the user of your northeastern email account as mentioned previously**.

Provide a **screenshot-2.1** showing all this new users into the /etc/passwd file

**Task 3 - Create a new group of users**

Create a new group named superusers, and
include your_nu_name_admin1 and your_nu_name_admin2 users into the superusers group.

**Task 4 - (Access Control) Control access to a folder directory in the system**

Create a directory in the your_nu_name_admin1's ${HOME} named justforyour_nu_name and setup the permissions required to make it accessible (RWX) by any user of
the superusers group, but only read (R) permissions for the user your_nu_name_regular

Provide a **screenshot-2.2** that shows permissions configuration in folder justforyour_nu_name.

**Task 5 - (Access Control) Switch user and create a file**

Become user your_nu_name_admin1 in your system and create a simple text-file
named your_nu_name_file  into the justforyour_nu_name folder. That file should be a text file with the following content:

CY5001 <semester> <year>
"your northeastern email address"
"your full name"

For example:

smith.m@northeastern.edu
Mary Smith

If you need to be sure about what user are you using type whoami.

Provide a **screenshot-2.3** that shows permissions for your_nu_name_file

**Task 6 - (Access Control) Read a different user's file**.

Now, become user your_nu_name_regular in your system and try to read `/home/your_nu_name_admin1/justforyour_nu_name/your_nu_name_file'.

Provide a **screenshot-2.4** that shows the result that you obtain and an explanation of it.

**Task 7 - (Access Control) Write a simple text file**

Keep being your_nu_name_regular and try to write a simple text-file (any content) into the justforyour_nu_name folder.

Provide a **screenshot-2.5** that shows the result that you obtain and an explanation of it.

**Task 8 - (Access Control) Try to delete a file**

Keep being your_nu_name_regular and try to delete the file `/home/your_nu_name_admin1/justforyour_nu_name/your_nu_name_file'.

Provide a **screenshot-2.6** that shows the result that you obtain and an explanation of it.

**Task 9 - (Auditing) Check audit logs**

Comeback to your root in your system and look for the "login events" into the appropriate file of the logs directory.

Provide a **screenshot-2.7** with the specific event into the appropriate log file.

**Task 10 - (Check known vulnerabilities) Install security updates**

Perform an operating system update, make sure that with your update command, all security patches are covered.

Provide a **screenshot-2.8** of the appropriate command you have used

**Task 11 - (Confidentiality) Install eCryptfs in Linux and create an encrypted folder directory**

Continue with your own linux user (not the new ones you created as part of this assignment), which possess sudo privileges, and install eCryptfs.

$ sudo apt install ecryptfs-utils -y

- Now, create a directory called CY5001 under your home directory using syntax *mkdir ~/CY5001*.
- Now, We need to encrypt this directory. It can be done with below syntax,

$ sudo mount -t ecryptfs ~/CY5001 ~/CY5001

Provide a passphrase that acts as encryption key and details specified as below,

- **Cipher:** aes
- **Key bytes :** 32
  - AES-256 encryption algorithm is one of the very strongest cryptographic available and it's strength increases exponential to it's key size. Hence, We have selected aes cipher and 32 bytes as key size.
- **Enable plaintext passthrough :** n
  - Enabling this option allows non-encrypted files to be used inside the mount. As we don't want to use non-encrypted files, You can disable this option by selecting no.
- **Enable filename encryption:** n
  - Enabling this option will encrypt the filenames also along with their contents. Disabling it will only encrypt the file contents.
- **Would you like to proceed with the mount? :** yes
- **Would you like to append sig to avoid warning in future? :** yes
  - This option will add the encryption signature to the key ring there by not giving any errors when you are trying to access the files again.

**Task 12 - (Confidentiality) Encrypt a file stored into the CY5001 directory,**

$ sudo nano ~/CY5001/YourLastName.txt

- Enter file text as mentioned below,

Your name : NUID
CY5001 <Semester> <year>

**Verifying Encryption**

- Enter below command and take **SCREENSHOT-2.9** of the output.

$ cat ~/CY5001/sample.txt

- You will see that the document is unencrypted and you can view the content. It is due to the reason that the directory is mounted at this time. Remember that as long as the directory is mounted, you will be able to see the content unless you unmount it.

**Task 13 - Unmount your encrypted directory**

Now let's try unmounting the directory using syntax below and view file. Take **SCREENSHOT-2.10** after running below commands

$ sudo umount ~/CY5001
$ cat ~/CY5001/sample.txt

Now, switch to ***root*** and repeat the last commands with the superuser privilege.

$ sudo su
$ cat **YOUROWNLINUXUSER**/CY5001/sample.txt

Take **SCREENSHOT-2.11** after running the previous commands.

**Note:** If you wanted to access this directory again, You need to mount it again to decrypt the directory.

**Task 14 - Stop asciinema recording**

This is the last task. Type **exit** and then don't forget to **press enter**.

- Also verify that you obtain a similar output.

$ exit

~ Asciicast recording finished.

~ Press <Enter> to upload, <Ctrl-C> to cancel.

View the recording at:

https://asciinema.org/a/v7YOamXMkkgV9kbKvvSy6FWRFj

This installation of asciinema recorder has not been linked to any

asciinema.org account.

All unclaimed recordings (from unknown installations like this one)

are automatically archived 7 days after upload.

If you want to preserve all recordings made on this machine, connect

this installation with asciinema.org account

by opening the following link:

You need now to export the cast file that you have created, from your Linux VM to your host computer (laptop), so you can submit the file through Canvas. To do so, we highly recommend you to use `scp` command line tool, because you will need to use it in many other assignments. In the first module of the class you can find a video about how to do scp.

Sources of information

- Linux Basics: [Linux Command Line for Beginners](https://linuxjourney.com/)
- User Management: [User Management](https://documentation.ubuntu.com/server/how-to/security/user-management/)
- Encryption: [eCryptfs Documentation](https://wiki.archlinux.org/title/ECryptfs)

Deliverables

Deliverable for this assignment will consist in to files:

1. Asciinema YourName.cast file you have created along the Phase 2 process
2. A pdf document report YourName.pdf with screenshots and explanations of the process you have followed.

Grading

Please see here the rubric for the assignment.

| **Concept** | **Points** |
|---|---|
| Phase 1. Tasks 1 & 2 | 1 |
| Phase 2. Tasks 1 & 14: Successful .cast file creation and submission | 2 |
| Phase 2. Tasks 2 & 3: Successful user and Group tasks | 1 |
| Phase 2. Tasks 4 - 8: Successful access control | 2 |
| Phase 2. Task 9: Successful identification of logs | 1 |
| Phase 2. Task 10 Successful security update. | 1 |
| Phase 2. Task 11 to 13. Successful encryption/decryption (confidentiality) | 2 |